SFU in BDCOM S9500 Chassis

What is an SFU Card?

- The SFU (**Switch Fabric Unit)** provides the high-speed interconnection backbone within the switch, enabling data forwarding between line cards.
- SFU handles tasks like packet forwarding, load balancing, and redundancy for fault tolerance.

Figure: BDCOM Chassis S9514 back side

There are total 6 slots in BDCOM S9500 Series backside for the SFU. However, it can be used maximum 4 SFU and rest of 2 are expandable in future.

SFU Models:

There are below 3 models of SFU with the corresponding chassis model based on the physical size

SFP Model	Chassis Model	Chassis Size
LS9506-SFU	S9506	11U size with total 6 slots (4 Business+2 MCU slots)
LS9510-SFU-II	S9510	14U size with total 6 slots (8 Business+2 MCU slots)
LS9514-SFU-II	S9514	18U size with total 6 slots (12 Business+2 MCU slots)

S9514 Switching Fabric Relationship (In slot 5~8)

line card	1*SFU	2*SFU	3*SFU	4*SFU
LS95-48GT (48 GE TX)	40Gbps	wire-speed	wire-speed	wire-speed
LS95-48GS (48 GE SFP)	40Gbps	wire-speed	wire-speed	wire-speed
LS95-48GT-POE (48 GE PoE)	40Gbps	wire-speed	wire-speed	wire-speed
LS95-48TS (48 GE 10G SFP+)	160Gbps	320Gbps	wire-speed	wire-speed
LS95-8QS (8 40GE QSFP+)	80Gbps	160Gbps	240Gbps	wire-speed
LS95-4CF (4 100GE QSFP28)	120Gbps	240Gbps	360Gbps	wire-speed

S9514 Switching Fabric Relationship (In other slots)

line card	1*SFU	2*SFU	3*SFU	4*SFU
LS95-48GT (48 GE TX)	40Gbps	wire-speed	wire-speed	wire-speed
LS95-48GS (48 GE SFP)	40Gbps	wire-speed	wire-speed	wire-speed
LS95-48GT-POE (48 GE PoE)	40Gbps	wire-speed	wire-speed	wire-speed
LS95-48TS (48 GE 10G SFP+)	80Gbps	160Gbps	240Gbps	320Gbps
LS95-8QS (8 40GE QSFP+)	80Gbps	160Gbps	240Gbps	wire-speed
LS95-4CF (4 100GE QSFP28)	80Gbps	160Gbps	240Gbps	320Gbps

Example:

- If insert 1*48GT and 1*48GS, to get wire-speed, need two SFUs. If insert only 1*48GT and 1 SFU then the bandwidth speed will be 40Gbps
- If insert 1*48GT and 1*48TS, to get wire-speed, need three SFUs
- If insert 1*48GT, 1*48GT-POE and 1*48TS, to get wire-speed, need three SFUs
- If insert 1 or more 4CF, need four SFUs
- If insert 4*48TS or the maximum 12*48TS, to get the wire speed need four SFUs

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